

Polynomial Stabilization of Highly Nonlinear Stochastic Systems with Pantograph Delay and Markovian Switching*

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DOI:

Received: x x 20xx / Revised: x x 20xx

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Abstract In this paper, we address the problem of almost sure polynomial stabilization for a class of highly nonlinear stochastic systems via sampled-data feedback. The considered systems fall within a general framework that includes two key features: (a) continuous-time irreducible Markov chain- we introduce a continuous-time irreducible Markov chain to describe systems that can undergo sudden alterations in their parameters and structures. This flexibility allows us to model real-world scenarios more accurately; (b) diffusion and drift coefficients with polynomial growth - unlike existing literature that primarily focuses on systems with bounded delays, we investigate the stabilization conditions for highly nonlinear stochastic systems with pantograph delay, an unbounded delay. Specifically, we analyze systems where the diffusion and drift coefficients satisfy a polynomial growth condition. To achieve our goal, we employ M-matrix theory and Lyapunov functions as basic tools. Our main results establish that the system can attain almost sure polynomial stabilization through a subtly and innovatively designed sampled-data feedback. We validate our theoretical findings with numerical simulations, demonstrating the effectiveness of our approach. This work contributes to the understanding of stabilization in highly nonlinear stochastic systems, particularly those with unbounded delays, and broadens the practical applicability of stochastic modeling.

Keywords Highly nonlinear stochastic systems, Markov chain, Almost sure polynomial stabilization, Pantograph delay.

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*This research was supported by the National Key R & D Program of China under Grants 2023YFE0126800, the NNSF of China under Grants 12371448 and 12401585, and the National Science Foundation of Guangdong Province of China under Grant 2024A1515011542.

◇ This paper was recommended for publication by Editor XX.

1 Introduction

Stochastic Differential Delay Equations (SDDEs) have garnered increasing attention within the scientific community. These equations capture the interplay of random factors and time delays, both of which are prevalent in numerous real-world systems^[1–3]. Moreover, many of these systems exhibit the intriguing property of sudden alterations in their parameters and structures. To enhance the fidelity of modeling such phenomena, researchers have introduced Markov chains into the framework, resulting in hybrid SDDEs. These hybrid equations provide a powerful tool for simulating complex dynamics, especially when dealing with systems in biology, mechanics, finance, and other fields. The incorporation of Markovian switching allows us to represent the inherent variability and adaptability observed in real-world processes. In literature, SDDEs with Markovian switching serve as a versatile mathematical framework for understanding and analyzing dynamic systems across diverse domains.

The study of stability and stabilization remains a crucial concern in the context of hybrid SDDEs. A substantial body of literature exists on the stabilization of hybrid SDDEs, including works by authors such as [4–13]. However, their theory is only applicable to hybrid SDDEs where the time delay is bounded, which is somewhat restrictive for unbounded delay stochastic systems. In recent years, a special case of SDDEs has garnered attention: Pantograph Stochastic Differential Equations (PSDEs), which is a stochastic differential equation with unbounded time delay. The current behavior of PSDEs is related to the current state and proportional past states. Therefore, PSDEs have the advantages of controllability and predictability, which have garnered widespread attention in engineering and control disciplines, including in the study of dynamical systems and the absorption of light by interstellar matter^[15, 16]. Real-world scenarios often involve specialized instances of PSDEs. Researchers have delved into the exponential and polynomial stability of PSDEs, leading to a rich body of work. Notable contributions include studies by Zhou et al.^[17] (focusing on exponential stability), Hu et al.^[18] and Milošević et al.^[19] (exploring polynomial stability) for PSDEs with Markovian switching. Furthermore, Mao et al.^[20] introduced the concept of the Lévy process and investigated the asymptotic stability of PSDEs. It is essential to note that these studies often impose conditions on the system coefficients, ensuring linear growth or make assumptions about the diffusion operator LV .

In reality, many nonlinear systems exhibit a high degree of complexity, and their coefficients no longer meet traditional linear growth conditions. To meet practical complexity requirements, Fei et al.^[21] introduced highly nonlinear systems. Specifically, the coefficients of these systems satisfy the polynomial growth condition, a less stringent requirement than the linear growth condition. This has led to significant advancements in highly nonlinear hybrid SDDEs^[4–6, 22–27]. The existing literature focuses on hybrid SDEs with bounded time delay, rendering it inapplicable to long or unbounded delay stochastic systems. The pantograph delay includes the case of unbounded delay. Therefore, studying the stability of stochastic systems with high nonlinearity and proportional delay is of both theoretical and practical significance. Moreover, polynomial stability can be effectively addressed using algorithms, which is crucial for many practical applications such as networks and economic models. Thus, the concept of polynomial

stability has applications in various fields, including automatic control, signal processing, economics, and biology. To reduce control costs, this paper focuses on designing an appropriate controller based on discrete-time observations to stabilize hybrid PSDEs with highly nonlinear coefficients, including H_∞ stabilization, asymptotic stabilization, and polynomial stabilization.

Motivated by the above discussions, we examine a highly nonlinear hybrid PSDE

$$dy(t) = f(y(t), y(\varrho t), \sigma(t), t)dt + g(y(t), y(\varrho t), \sigma(t), t)dB(t) \quad (1)$$

where $0 < \varrho < 1$ is the pantograph coefficient, $f : \mathbb{R}^n \times \mathbb{R}^n \times S \times \mathbb{R}_+ \rightarrow \mathbb{R}^n$, and $g : \mathbb{R}^n \times \mathbb{R}^n \times S \times \mathbb{R}_+ \rightarrow \mathbb{R}^{n \times m}$ are Borel measurable functions and satisfy polynomial growth conditions. Here $B(t)$ is a Brownian motion, $S = \{1, 2, \dots, N\}$ is a state space of the Markov chain $\sigma(t)$, the state $y(t)$ takes values in $\mathbb{R}^n$. Let $B(t)$ and $\sigma(t)$ be independent of each other. The highly nonlinear hybrid PSDE often cannot achieve stability directly. Therefore, a suitable control strategy is proposed to ensure the desirable stability^[25–33]. Note that the realization of continuous-time feedback controllers requires continuous observation of $y(t), \forall t \geq t_0$. In practical applications, observations are often performed only at discrete time intervals $t_0, t_0 + \tau, \dots$, due to various reasons. Therefore, we will devise a controller based on discrete-time observations $u(y(t_0 + [\frac{t-t_0}{\tau}]\tau), \sigma(t), t)$.

We would like to highlight several key points in our study:

1. Our work diverges from existing literature^[17, 18, 20] by focusing on the stabilization of a specific class of highly nonlinear SDEs that exhibit pantograph delay and are influenced by a Markov chain. Furthermore, we delve into the convergence rate at which the solution approaches zero, analyzing its convergence rate by examining both polynomial stabilization and almost sure polynomial stabilization of the controlled PSDE.
2. In contrast to previous studies on the boundedness of the state of PSDEs^[34, 35], we propose a novel criterion, which is Hypothesis 3. This criterion is unique in that it does not rely on the transition matrix but is directly influenced by the structure of the diffusion and drift coefficients of the PSDEs.

The paper is organised as follows. In Section 2, the preliminaries on the highly nonlinear hybrid PSDEs are given. On this basis, we design a sampled-data controller and impose some standing hypotheses which guarantee the finiteness and the boundedness of the unique solution of the controlled system. Subsequently, an appropriate control function is designed, and it then is shown that such a sampled-data feedback control can achieve polynomial stabilization of the given PSDE in Section 3. Our theory is illustrated by an example in Section 4 while the paper is concluded in Section 5.

The following notations are employed in this paper. We use $|\cdot|$ to represent Euclidean norm in $\mathbb{R}^n$. A^T denotes the transpose of a matrix or vector. Let the norm of a matrix A be denoted as $\|A\| = \sup\{|Ay| : |y| = 1\}$. Let $t \geq t_0 > 0$, $C([\varrho t_0, t_0]; \mathbb{R}^n)$ be the space of continuous functions ψ mapping the interval $[\varrho t_0, t_0] \rightarrow \mathbb{R}^n$ with the norm $\|\psi\| = \sup_{\varrho t_0 \leq \theta \leq t_0} |\psi(\theta)|$. Let $l > 0$, and denote by $L_{\mathcal{F}_t}^l([\varrho t, t]; \mathbb{R}^n)$ the family of $\mathcal{F}_t$ -measurable $C([\varrho t, t]; \mathbb{R}^n)$ -valued random

variables ψ so that $\mathbb{E}\|\psi\|^l < \infty$. Let $C^{2,1}(\mathbb{R}^n \times S \times \mathbb{R}_+; \mathbb{R})$ be the family of all continuously twice differentiable and real-valued functions $V(y, i, t): \mathbb{R}^n \times S \times \mathbb{R}_+$. For function $V \in C^{2,1}$, we set $V_t = \frac{\partial V}{\partial t}$, $V_y = (\frac{\partial V}{\partial y_1}, \dots, \frac{\partial V}{\partial y_n})$ and $V_{yy} = (\frac{\partial^2 V}{\partial y_i \partial y_j})_{n \times n}$. Let $\gamma_{ij} \geq 0$ be the transition rate from i to j if $i \neq j$, and $\Gamma = (\gamma_{ij})_{N \times N}$ be the generator of $\{\sigma(t), t \geq 0\}$. Let $a, b \in \mathbb{R}$, $a \vee b = \max\{a, b\}$ and $a \wedge b = \min\{a, b\}$.

2 The finiteness and the boundedness of the global solution

Due to the unbounded delay of PSDE (1), the highly nonlinear hybrid PSDE (1) may not achieve stability on its own, in which case we need to design a suitable controller to achieve stability. Before proving the stability of the system, we need to do some preparatory work. We will prove the finiteness of the moment of the solution for subsequent use in studying polynomial stabilization of the controlled system in the mean square sense, as well as the boundedness of the moment of the solution for studying almost sure polynomial stabilization of the controlled system.

Suppose that the underlying system is the PSDE (1) on $t \geq t_0$ with the initial data

$$\{y(t) : \varrho t_0 \leq t \leq t_0\} = \zeta \in L^l_{\mathcal{F}_t}([t_0, t_0]; \mathbb{R}^n) \text{ and } \sigma(t_0) = i_0, \quad (2)$$

where $0 < \varrho < 1$. In the event that the given PSDE (1) cannot directly achieve stability, we devise an appropriate sampled-data controller to promote the PSDE

$$dy(t) = [f(y(t), y(\varrho t), \sigma(t), t) + u(y(\delta_t), \sigma(t), t)]dt + g(y(t), y(\varrho t), \sigma(t), t)dB(t) \quad (3)$$

to be stable, where $\delta_t = t_0 + [\frac{t-t_0}{\tau}]\tau$. We can see that the controller contains time delay which is variable and bounded. Then, we can establish a function $\pi : [t_0, \infty] \rightarrow [0, \tau]$ to describe the time delay as

$$\pi(t) = t - (t_0 + i\tau) \quad \text{for } t_0 + i\tau \leq t < t_0 + (i+1)\tau, i = 0, 1, 2, \dots$$

Next, we will reformulate the controlled system (3) as

$$\begin{aligned} dy(t) &= [f(y(t), y(\varrho t), \sigma(t), t) + u(y(t - \pi(t)), \sigma(t), t)]dt \\ &\quad + g(y(t), y(\varrho t), \sigma(t), t)dB(t). \end{aligned} \quad (4)$$

For convenience, we assume $\varrho t_0 < t_0 - \tau$.

We assume the control function u satisfies the Lipschitz condition, that is

$$|u(y, i, t) - u(\hat{y}, i, t)| \leq \iota |y - \hat{y}| \quad (5)$$

for $\forall t \geq t_0, i \in S$, and $y, \hat{y} \in \mathbb{R}^n$, where $\iota > 0$. Furthermore, we demand that $u(0, i, t) \equiv 0$. This condition is common in PID and linear feedback control and is applied to [4, 5]. The specific design of the control function will be placed in the next section. Next, we will prove that the moment of the solution of the controlled system is finite and bounded when the control function u satisfies the condition (5).

Hypothesis 1 (Highly nonlinearity) Assume the diffusion and drift coefficient are locally Lipschitz continuous, and satisfy polynomial growth condition, that is

$$\begin{aligned} |f(y, \hat{y}, i, t)| &\leq L(|y| + |\hat{y}| + |y|^{l_1} + |\hat{y}|^{l_2}), \\ |g(y, \hat{y}, i, t)| &\leq L(|y| + |\hat{y}| + |y|^{l_3} + |\hat{y}|^{l_4}) \end{aligned} \quad (6)$$

for $\forall (y, \hat{y}, i, t) \in \mathbb{R}^n \times \mathbb{R}^n \times S \times \mathbb{R}_+$, where $L > 0$, $l_1 > 1$, $l_i \geq 1$ ($i = 2, 3, 4$).

To prove the finiteness of the solution of the system, we need to propose the following hypotheses.

Hypothesis 2 (Khasminskii-type condition) There exist some constants $q, r, \alpha_i > 0$ ($i = 1, 2, \dots, 4$) with $\alpha_3 > \alpha_4$ such that

$$q \geq 2(l_1 \vee l_2 \vee l_3 \vee l_4) - l_1 + 1, \quad (7)$$

$$r > q + l_1 - 1, \quad (8)$$

$$l > q + r - 2,$$

and for $\forall (y, \hat{y}, i, t) \in \mathbb{R}^n \times \mathbb{R}^n \times S \times \mathbb{R}_+$,

$$y^T f(y, \hat{y}, i, t) + \frac{r-1}{2} |g(y, \hat{y}, i, t)|^2 \leq \alpha_1 |y|^2 + \alpha_2 \varrho |\hat{y}|^2 - \alpha_3 |y|^q + \alpha_4 \varrho |\hat{y}|^q. \quad (9)$$

Remark 1 When there is no delay in the system, condition (9) is the same as the condition proposed in [4]. Due to the pantograph delay of the system, the condition (9) we propose is a broader scenario.

Lemma 2.1 Under (5), Hypotheses 1 and 2, PSDE (4) with the initial data

$$\{y(t) : \varrho t_0 \leq t \leq t_0\} = \zeta \in L_{\mathcal{F}_t}^r([\varrho t_0, t_0]; \mathbb{R}^n) \text{ and } \sigma(t_0) = i_0, \quad (10)$$

has a unique global solution $y(t)$ on $t \in [t_0, \infty)$ that satisfies

$$\mathbb{E}|y(t)|^{\tilde{r}} < \infty, \quad (11)$$

where $\tilde{r} = r - \varepsilon$, and $\varepsilon > 0$ is a sufficiently small constant.

Proof To facilitate comprehension of the proof, we will break it down into three steps.

Step 1. Select a function $\tilde{U}$ as

$$\tilde{U}(y) = |y|^r, \quad y \in \mathbb{R}^n. \quad (12)$$

Applying the Itô formula, then

$$\begin{aligned} d\tilde{U}(y(t)) &= L\tilde{U}(y(t), y(\varrho t), y(t - \pi(t)), \sigma(t), t)dt \\ &\quad + r|y(t)|^{r-2} y^T(t) g(y(t), y(\varrho t), \sigma(t), t) dB(t), \end{aligned}$$

where $L\tilde{U} : \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n \times S \times \mathbb{R}_+ \rightarrow \mathbb{R}$ is a function on the PSDE (4), and it is given by

$$\begin{aligned} L\tilde{U}(y, \hat{y}, z, i, t) &= r|y|^{r-2}y^T[f(y, \hat{y}, i, t) + u(z, i, t)] + \frac{r}{2}|y|^{r-2}|g(y, \hat{y}, i, t)|^2 \\ &\quad + \frac{r(r-2)}{2}|y|^{r-4}|y^T g(y, \hat{y}, i, t)|^2. \end{aligned} \quad (13)$$

Then, by Hypothesis 2, the Lipschitz condition (5) and Cauchy inequality,

$$L\tilde{U}(y, \hat{y}, z, i, t) \leq r|y|^{r-2}[\alpha_1|y|^2 + \alpha_2\varrho|\hat{y}|^2 - \alpha_3|y|^q + \alpha_4\varrho|\hat{y}|^q + \iota|y||z|]. \quad (14)$$

According to the Young's inequality, then

$$\begin{aligned} |y|^{r-2}|\hat{y}|^2 &\leq \frac{r-2}{r}|y|^r + \frac{2}{r}|\hat{y}|^r, \\ |y|^{r-2}|\hat{y}|^q &\leq \frac{r-2}{q+r-2}|y|^{q+r-2} + \frac{q}{q+r-2}|\hat{y}|^{q+r-2}, \\ |y|^{r-1}|z| &\leq \frac{r-1}{r}|y|^r + \frac{1}{r}|z|^r. \end{aligned}$$

According to (14), we get

$$\begin{aligned} L\tilde{U}(y, \hat{y}, z, i, t) &\leq \beta_1|y|^r + 2\alpha_2\varrho|\hat{y}|^r - (r\alpha_3 - \frac{r\alpha_4\varrho(r-2)}{q+r-2})|y|^{q+r-2} \\ &\quad + \frac{r\alpha_4q}{q+r-2}\varrho|\hat{y}|^{q+r-2} + \iota|z|^r, \end{aligned} \quad (15)$$

where $\beta_1 = r\alpha_1 + (r-2)\alpha_2\varrho + \iota(r-1)$.

Step 2. According to the result in [20], the local Lipschitz condition guarantees that the hybrid PSDE (4) with the initial data (10) has a unique maximal local solution, denoted by $y(t)$ on $[t_0, e_\infty)$, where e_∞ is the explosion time. For each integer $k \geq \|\zeta\|$, define the stopping time

$$\xi_k = \inf\{t \in [t_0, e_\infty) : |y(t)| \geq k\}. \quad (16)$$

Apparently, ξ_k is non-decreasing and $\xi_k \rightarrow \xi_\infty \leq e_\infty$ a.s. Therefore we merely need to demonstrate $P(\xi_\infty = \infty) = 1$. Restrict $t \in [t_0, t_0 + \tau]$. Noting that $\varrho t_0 \leq \varrho t < t_0$, we conclude that $y(\varrho t)$ is known. Because of

$$t - \pi(t) = \begin{cases} t_0 & t_0 \leq t < t_0 + \tau, \\ t_0 + \tau & t = t_0 + \tau. \end{cases}$$

By the Itô formula and (13)-(15), we can obtain

$$\begin{aligned} \mathbb{E}|y(\xi_k \wedge t)|^r - \mathbb{E}|y(t_0)|^r &= \mathbb{E} \int_{t_0}^{\xi_k \wedge t} L\tilde{U}(y(s), y(\varrho s), y(s - \pi(s)), \sigma(s), s) ds \\ &\leq \mathbb{E} \int_{t_0}^{\xi_k \wedge t} \beta_1|y(s)|^r ds + \mathbb{E} \int_{t_0}^{\xi_k \wedge t} 2\alpha_2\varrho|y(\varrho s)|^r ds + \mathbb{E} \int_{t_0}^{\xi_k \wedge t} \iota|y(s - \pi(s))|^r ds \\ &\quad - \mathbb{E} \int_{t_0}^{\xi_k \wedge t} (r\alpha_3 - \frac{r\alpha_4\varrho(r-2)}{q+r-2})|y(s)|^{q+r-2} ds + \mathbb{E} \int_{t_0}^{\xi_k \wedge t} \frac{r\alpha_4q}{q+r-2}\varrho|y(\varrho s)|^{q+r-2} ds, \end{aligned} \quad (17)$$

Clearly derived from the expression of the control function, we can conclude that the value of $y(s - \pi(s))$ remains $y(t_0)$ during $[t_0, t_0 + \tau)$. Therefore, we can obtain $\mathbb{E} \int_{t_0}^{\xi_k \wedge t} \iota |y(s - \pi(s))|^r ds \leq \iota \tau \mathbb{E} |y(t_0)|^r$. Additionally, it is noticed that for any $t \geq t_0$,

$$\begin{aligned} \mathbb{E} \int_{t_0}^t \varrho |y(\varrho s)|^r ds &\leq (1 - \varrho) t_0 \mathbb{E} \|\zeta\|^r + \mathbb{E} \int_{t_0}^t |y(s)|^r ds, \\ \mathbb{E} \int_{t_0}^t \varrho |y(\varrho s)|^{q+r-2} ds &\leq (1 - \varrho) t_0 \mathbb{E} \|\zeta\|^{q+r-2} + \mathbb{E} \int_{t_0}^t |y(s)|^{q+r-2} ds. \end{aligned}$$

Consequently, based on (17), we can deduce that for any $t \in [t_0, t_0 + \tau]$,

$$\begin{aligned} &\mathbb{E} |y(\xi_k \wedge t)|^r + \beta_2 \mathbb{E} \int_{t_0}^{\xi_k \wedge t} |y(s)|^{q+r-2} ds \\ &\leq \mathbb{E} |y(t_0)|^r + \alpha_5 + (\beta_1 + 2\alpha_2) \mathbb{E} \int_{t_0}^{\xi_k \wedge t} |y(s)|^r ds, \end{aligned} \quad (18)$$

where $\beta_2 = r\alpha_3 - \frac{r\alpha_4\varrho(r-2)}{q+r-2} - \frac{r\alpha_4q}{q+r-2} > 0$, $\alpha_5 = (1 - \varrho)t_0[2\alpha_2\mathbb{E}\|\zeta\|^r + \frac{r\alpha_4q}{q+r-2}\mathbb{E}\|\zeta\|^{q+r-2}] + \iota\tau\mathbb{E}|y(t_0)|^r$, which is finite clearly. It follows from (18) that

$$\mathbb{E} |y(\xi_k \wedge t)|^r \leq \mathbb{E} |y(t_0)|^r + \alpha_5 + (\beta_1 + 2\alpha_2) \mathbb{E} \int_{t_0}^{\xi_k \wedge t} |y(s)|^r ds.$$

An application of the well-known Gronwall inequality yields

$$\mathbb{E} |y(\xi_k \wedge t)|^r \leq (\mathbb{E} |y(t_0)|^r + \alpha_5) e^{(\beta_1 + 2\alpha_2)\tau} =: \alpha_6 \quad (19)$$

for all $t \in [t_0, t_0 + \tau]$, where throughout this paper we use $=:$ to stand for ‘denoted by’. This implies $k^r P(\xi_k \leq (t_0 + \tau)) \leq \mathbb{E} |y(\xi_k \wedge (t_0 + \tau))|^r \leq \alpha_6$. Letting $k \rightarrow \infty$, we can have

$$P(\xi_\infty > t_0 + \tau) = 1. \quad (20)$$

For the given sufficient small ε , we know $\{|y(\xi_k \wedge t)|^{r-\frac{\varepsilon}{2}}\}_{k \geq 1}$ is uniformly integrable by (19) for each $t \in [t_0, t_0 + \tau]$. Letting $k \rightarrow \infty$ in (19), we have

$$\mathbb{E} |y(t)|^{\tilde{r}_1} \leq \alpha_6, \forall t \in [t_0, t_0 + \tau], \quad (21)$$

where $\tilde{r}_1 = r - \frac{\varepsilon}{2}$.

Step 3. We now show $P(\xi_\infty > t_0 + n\tau) = 1$ and the boundedness of $\mathbb{E}[|y(t)|^{\tilde{r}_n}]$ on $[t_0, t_0 + n\tau]$ for each n , where $\tilde{r}_n = \tilde{r}_{n-1} - \frac{\varepsilon}{2^n}$, $\tilde{r}_0 = r$. The case of $n = 1$ has been proved in the last step. Now suppose $P(\xi_\infty > t_0 + (n-1)\tau) = 1$, and $\mathbb{E}[|y(t)|^{\tilde{r}_{n-1}}]$ is bounded on $[t_0, t_0 + (n-1)\tau]$. Restrict $t \in [t_0, t_0 + n\tau]$. We also observe that $\varrho t_0 \leq \varrho t < t_0 + (n-1)\tau$ whenever $t \in [t_0, t_0 + n\tau]$. It is easy to see that (17) still holds for $t \in [t_0, t_0 + n\tau]$, $r = \tilde{r}_{n-1}$. On the other hand,

$$t - \pi(t) = \begin{cases} t_0 & t_0 \leq t < t_0 + \tau, \\ t_0 + m\tau & t_0 + m\tau \leq t < t_0 + (m+1)\tau, \ 1 \leq m \leq n-1, \\ t_0 + n\tau & t = t_0 + n\tau. \end{cases} \quad (22)$$

In the same way as (18) was proven, we can show that

$$\begin{aligned} & \mathbb{E}|y(\xi_k \wedge t)|^{\tilde{r}_{n-1}} + \beta_2 \mathbb{E} \int_{t_0}^{\xi_k \wedge t} |y(s)|^{q+\tilde{r}_{n-1}-2} ds \\ & \leq |\zeta(t_0)|^{\tilde{r}_{n-1}} + \alpha_8 + (\beta_1 + 2\alpha_2) \mathbb{E} \int_{t_0}^{\xi_k \wedge t} |y(s)|^{\tilde{r}_{n-1}} ds, \end{aligned} \quad (23)$$

where $\alpha_8 = (1 - \varrho)t_0[2\alpha_2\mathbb{E}\|\zeta\|^{\tilde{r}_{n-1}} + \frac{\tilde{r}_{n-1}\alpha_4q}{q+\tilde{r}_{n-1}-2}\mathbb{E}\|\zeta\|^{q+\tilde{r}_{n-1}-2}] + \iota\tau\sum_{m=0}^{n-1}\mathbb{E}|y(t_0+m\tau)|^{\tilde{r}_{n-1}}$, which is finite by induction assumptions. Then $\mathbb{E}|y(\xi_k \wedge t)|^{\tilde{r}_{n-1}}$ is bounded by Gronwall inequality. Hence $|y(\xi_k \wedge t)|^{\tilde{r}_n}$ is uniformly integrable. By letting k tend to infinity, we have

$$\mathbb{E}|y(t)|^{\tilde{r}_n} < \infty, \forall t \in [t_0, t_0 + n\tau].$$

In a similar way to step 2, we can show $P(\xi_\infty > t_0 + n\tau) = 1$. Then we obtain that PSDE (4) has a global solution. Noticing

$$\mathbb{E}|y(t)|^{\tilde{r}} \leq \mathbb{E}|y(t)|^{\tilde{r}_n} < \infty,$$

when $t \in [t_0, t_0 + n\tau]$, we obtain the results.

The proof is therefore complete.

We have proven the finiteness of the moments of the solution for subsequent use in studying polynomial stabilization of the controlled PSDE (4) in the mean square sense. Next, we also need to prove the boundedness of the moments of the solution in order to study almost sure polynomial stabilization of the controlled PSDE (4). For this purpose, we need to strengthen Hypothesis 2 slightly.

The following hypotheses strengthen the constraint on the coefficients of PSDE (4), and we provide a lemma to prove that the solution of PSDE (4) is bounded under these hypotheses.

Hypothesis 3 We replace (9) in Hypothesis 2 with the following condition

$$y^T f(y, \hat{y}, i, t) + \frac{r-1}{2} |g(y, \hat{y}, i, t)|^2 \leq \alpha_1 |y|^2 + \alpha_2 \varrho e^{-\theta(t)} |\hat{y}|^2 - \alpha_3 |y|^q + \alpha_4 \varrho e^{-\theta(t)} |\hat{y}|^q, \quad (24)$$

where $\theta(t) = (1 - \varrho)t$, for $\forall (y, \hat{y}, i, t) \in \mathbb{R}^n \times \mathbb{R}^n \times S \times \mathbb{R}_+$.

Lemma 2.2 Under (5), Hypotheses 1 and 3, the solution of the PSDE (4) initialized with the provided data (10) meets

$$\sup_{t_0 \leq t < \infty} \mathbb{E}|y(t)|^{\tilde{r}} < \infty, \quad (25)$$

where $\tilde{r} = r - \delta$ for sufficient small δ .

Proof Let $1 > \varepsilon > 0$ be sufficiently small such that $\alpha_3 \geq \alpha_4 + \varepsilon$. Let ξ_k be given by (16).

By the Itô formula and Hypotheses 3, it is easy to show that for each k

$$\begin{aligned}
& e^{\varepsilon(t \wedge \xi_k)} \mathbb{E}|y(t \wedge \xi_k)|^r - e^{\varepsilon t_0} |\zeta(t_0)|^r \\
& \leq \mathbb{E} \int_{t_0}^{t \wedge \xi_k} e^{\varepsilon s} \left([\varepsilon + r\alpha_1 + (r-2)\alpha_2 \varrho e^{-\theta(s)} + \iota(r-1)] |y(s)|^r + \iota |y(s-\tau)|^r \right. \\
& \quad \left. + 2\alpha_2 \varrho e^{-\theta(s)} |y(\varrho s)|^r - (r\alpha_3 - \frac{r(r-2)\alpha_4 \varrho e^{-\theta(s)}}{q+r-2}) |y(s)|^{q+r-2} \right. \\
& \quad \left. + \frac{r\alpha_4 q \varrho e^{-\theta(s)}}{q+r-2} |y(\varrho s)|^{q+r-2} \right) ds. \tag{26}
\end{aligned}$$

Additionally, it is noticed that

$$\begin{aligned}
& \mathbb{E} \int_{t_0}^t e^{\varepsilon s} \varrho e^{-\theta(s)} |y(\varrho s)|^{q+r-2} ds \\
& = \mathbb{E} \int_{\varrho t_0}^{\varrho t} e^{\varepsilon \frac{\rho}{\varrho}} e^{-\theta(\frac{\rho}{\varrho})} |y(\rho)|^{q+r-2} d\rho \\
& \leq \mathbb{E} \int_{\varrho t_0}^t e^{\varepsilon \rho} e^{\varepsilon(\frac{\rho}{\varrho} - \rho) - \theta(\frac{\rho}{\varrho})} |y(\rho)|^{q+r-2} d\rho \\
& \leq \mathbb{E} \int_{\varrho t_0}^t e^{\varepsilon s} |y(s)|^{q+r-2} ds \\
& \leq \frac{e^{\varepsilon t_0} - e^{\varepsilon \varrho t_0}}{\varepsilon} \mathbb{E} \|\zeta\|^{q+r-2} + \mathbb{E} \int_{t_0}^t e^{\varepsilon s} |y(s)|^{q+r-2} ds.
\end{aligned}$$

Consequently, based on (26), we can deduce that

$$\begin{aligned}
& e^{\varepsilon(t \wedge \xi_k)} \mathbb{E}|y(t \wedge \xi_k)|^r - e^{\varepsilon t_0} |\zeta(t_0)|^r \\
& \leq C_1 + \mathbb{E} \int_{t_0}^{t \wedge \xi_k} e^{\varepsilon s} \left(\alpha_9 - (r\alpha_3 - \frac{r(r-2)\alpha_4 \varrho e^{-\theta(s)}}{q+r-2} - \frac{r\alpha_4 q}{q+r-2} - r\varepsilon) |y(s)|^{q+r-2} \right) ds,
\end{aligned}$$

where $C_1 = \frac{\varepsilon(1-\varrho)t_0}{\varepsilon} (2\alpha_2 \mathbb{E} \|\zeta\|^r + \frac{r\alpha_4 q}{q+r-2} \mathbb{E} \|\zeta\|^{q+r-2})$, $\alpha_9 = \sup_{u \geq 0} ([\varepsilon + r\alpha_1 + (r-2)\alpha_2 \varrho + \iota(r-1) + 2\alpha_2 + \iota] u^r - r\varepsilon u^{q+r-2})$. This implies

$$\mathbb{E}|y(t \wedge \xi_k)|^r \leq e^{\varepsilon t_0} |\zeta(t_0)|^r + C_1 + \frac{\alpha_9}{\varepsilon}$$

for all $t \geq t_0$. Noticing the right-handed side is independent of k , we know that $\{|y(\xi_k \wedge t)|^{r-\delta}\}_{k \geq 1}$ is uniformly integrable. By letting k tend to infinity, we arrive at assertion (25).

3 Polynomial stabilization

Below we give the formal definitions of p th moment polynomial stability and the almost sure polynomial stability.

Definition 3.1 ^[36] The trivial solution of equation (1) is said to be p th moment polynomially stable if

$$\limsup_{t \rightarrow \infty} \frac{\log \mathbb{E}|y(t)|^p}{\log(1+t)} < 0$$

for all initial data $\zeta \in L^r_{\mathcal{F}_t}([\varrho t_0, t_0]; \mathbb{R}^n)$. When $p = 2$, it is usually said to be polynomially stable in mean square.

Definition 3.2 ^[36] The trivial solution of equation (1) is said to be almost surely polynomially stable if

$$\limsup_{t \rightarrow \infty} \frac{\log |y(t)|}{\log(1+t)} < 0 \quad a.s.$$

for all initial data $\zeta \in L^r_{\mathcal{F}_t}([\varrho t_0, t_0]; \mathbb{R}^n)$.

Although we have just shown that the solution of the controlled system (4) is finite and bounded, we need to design the control function more carefully in order for the controlled system (4) to be polynomially stable, that is, the rate at which the solution tends to 0 is controlled by a polynomial with a negative exponent.

Assumption 3.1 Under Hypothesis 2, for any controller $u : \mathbb{R}^n \times S \times \mathbb{R}_+ \rightarrow \mathbb{R}^n$ satisfying (5), there exist $\tilde{a}_i, \hat{a}_i \in \mathbb{R}$, $\tilde{c}_i, \hat{c}_i \in \mathbb{R}_+$, and $\tilde{b}_i \geq 0, \hat{b}_i \geq 0, \tilde{d}_i \geq 0, \hat{d}_i \geq 0 (i \in S)$ such that the following inequalities hold

$$y^T[f(y, \hat{y}, i, t) + u(y, i, t)] + \frac{1}{2}|g(y, \hat{y}, i, t)|^2 \leq \tilde{a}_i|y|^2 + \tilde{b}_i\varrho|\hat{y}|^2 - \tilde{c}_i|y|^q + \tilde{d}_i\varrho|\hat{y}|^q, \quad (27)$$

$$y^T[f(y, \hat{y}, i, t) + u(y, i, t)] + \frac{l_1}{2}|g(y, \hat{y}, i, t)|^2 \leq \hat{a}_i|y|^2 + \hat{b}_i\varrho|\hat{y}|^2 - \hat{c}_i|y|^q + \hat{d}_i\varrho|\hat{y}|^q, \quad (28)$$

for $\forall (y, \hat{y}, i, t) \in \mathbb{R}^n \times \mathbb{R}^n \times S \times \mathbb{R}_+$; while let M_1, M_2 be nonsingular M-matrices,

$$M_1 =: -2\text{diag}(\tilde{a}_1, \dots, \tilde{a}_N) - \Gamma, \quad M_2 =: -(l_1 + 1)\text{diag}(\hat{a}_1, \dots, \hat{a}_N) - \Gamma, \quad (29)$$

and

$$1 > \gamma_1, \gamma_2 > \gamma_3, 1 > \gamma_4, \gamma_5 > \gamma_6, \quad (30)$$

where

$$\begin{aligned} \gamma_1 &= \max_{i \in S} 2\vartheta_i \tilde{b}_i, & \gamma_2 &= \min_{i \in S} 2\vartheta_i \tilde{c}_i, & \gamma_3 &= \max_{i \in S} 2\vartheta_i \tilde{d}_i, \\ \gamma_4 &= \max_{i \in S} (l_1 + 1)\bar{\vartheta}_i \hat{b}_i, & \gamma_5 &= \min_{i \in S} (l_1 + 1)\bar{\vartheta}_i \hat{c}_i, & \gamma_6 &= \max_{i \in S} (l_1 + 1)\bar{\vartheta}_i \hat{d}_i, \end{aligned}$$

in which

$$(\vartheta_1, \dots, \vartheta_N)^T = M_1^{-1}(1, \dots, 1)^T, \quad (\bar{\vartheta}_1, \dots, \bar{\vartheta}_N)^T = M_2^{-1}(1, \dots, 1)^T. \quad (31)$$

Assumption 3.1 is an enhanced version of Hypothesis 2, which adds a controller and satisfies the Khasminskii-type condition. Assumption 3.1, as well as Assumption 3.2 in the following, are very common in the study of high-nonlinearity systems with delays, as seen in references [4], [5], [6], and [12]. In the design of controllers, the M-matrix is a very important tool. Below, we will provide its definition.

Definition 3.3 A square matrix $A = (a_{ij})_{n \times n}$ is called a nonsingular M-matrix if A can be expressed in the form $A = sI - G$ with some $G \geq 0$ and $s > p(G)$, where I is the identity $n \times n$ matrix and $p(G)$ the spectral radius of G .

We should point out that all ϑ_i and $\bar{\vartheta}_i$ defined by (31) are positive as both M_1 and M_2 are nonsingular M-matrices. Regarding the theory on M-matrices we refer the reader to Mao and Yuan (2006, Section 2.6). The ϑ_i and $\bar{\vartheta}_i$ estimated by parameters $\tilde{a}_i, \hat{a}_i, \tilde{c}_i, \hat{c}_i, \tilde{b}_i, \hat{b}_i, \tilde{d}_i, \hat{d}_i$ are used for constructing the Lyapunov function (32). Parameters γ_i are used for proposing Assumption 3.2.

We define a real valued function U by

$$U(y, i) = \vartheta_i |y|^2 + \bar{\vartheta}_i |y|^{l_1+1}, (y, i) \in \mathbb{R}^n \times S, \quad (32)$$

and a function $\mathcal{L}U : \mathbb{R}^n \times \mathbb{R}^n \times S \times \mathbb{R}_+ \rightarrow \mathbb{R}$, which is expressed as follows.

$$\begin{aligned} & \mathcal{L}U(y, \hat{y}, i, t) \\ &= 2\vartheta_i \left[y^T [f(y, \hat{y}, i, t) + u(y, i, t)] + \frac{1}{2} |g(y, \hat{y}, i, t)|^2 \right] + \sum_{j=1}^N \gamma_{ij} (\vartheta_j |y|^2 + \bar{\vartheta}_j |y|^{l_1+1}) \\ &+ (l_1 + 1) \bar{\vartheta}_i \left[|y|^{l_1-1} y^T [f(y, \hat{y}, i, t) + u(y, i, t)] + \frac{1}{2} |y|^{l_1-1} |g(y, \hat{y}, i, t)|^2 \right. \\ &\left. + \frac{l_1 - 1}{2} |y|^{l_1-3} |y^T g(y, \hat{y}, i, t)|^2 \right]. \end{aligned} \quad (33)$$

By the (27), (28), (29) and (31), we can estimate (33) as

$$\begin{aligned} \mathcal{L}U(y, \hat{y}, i, t) &\leq 2\vartheta_i \left[\tilde{a}_i |y|^2 + \tilde{b}_i \varrho |\hat{y}|^2 - \tilde{c}_i |y|^q + \tilde{d}_i \varrho |\hat{y}|^q \right] \\ &\quad - (2\tilde{a}_i \vartheta_i + 1) |y|^2 - [(l_1 + 1) \hat{a}_i \bar{\vartheta}_i + 1] |y|^{l_1+1} \\ &\quad + (l_1 + 1) \bar{\vartheta}_i |y|^{l_1-1} \left[\hat{a}_i |y|^2 + \hat{b}_i \varrho |\hat{y}|^2 - \hat{c}_i |y|^q + \hat{d}_i \varrho |\hat{y}|^q \right]. \end{aligned} \quad (34)$$

By simplifying (34) and using (30), we can conclude that

$$\begin{aligned} \mathcal{L}U(y, \hat{y}, i, t) &\leq -|y|^2 + \gamma_1 \varrho |\hat{y}|^2 - \gamma_2 |y|^q + \gamma_3 \varrho |\hat{y}|^q \\ &\quad - |y|^{l_1+1} + \gamma_4 \varrho |\hat{y}|^2 |y|^{l_1-1} + \gamma_5 |y|^{q+l_1-1} + \gamma_6 \varrho |\hat{y}|^q |y|^{l_1-1}. \end{aligned} \quad (35)$$

Meanwhile, by Young's inequality, we can obtain

$$|\hat{y}|^2 |y|^{l_1-1} \leq \frac{2}{l_1 + 1} |\hat{y}|^{l_1+1} + \frac{l_1 - 1}{l_1 + 1} |y|^{l_1+1},$$

$$|\hat{y}|^q |y|^{l_1-1} \leq \frac{q}{l_1 + q - 1} |\hat{y}|^{l_1+q-1} + \frac{l_1 - 1}{l_1 + q - 1} |y|^{l_1+q-1}.$$

By the Young's inequality, (34) and (35), we can estimate (33) as

$$\begin{aligned} & \mathcal{L}U(y, \hat{y}, i, t) \\ &\leq -|y|^2 + \gamma_1 \varrho |\hat{y}|^2 - \gamma_2 |y|^q + \gamma_3 \varrho |\hat{y}|^q + \frac{2\gamma_4 \varrho}{l_1 + 1} |\hat{y}|^{l_1+1} - \left(1 - \frac{\gamma_4 (l_1 - 1) \varrho}{l_1 + 1} \right) |y|^{l_1+1} \\ &+ \frac{\gamma_6 q \varrho}{q + l_1 - 1} |\hat{y}|^{q+l_1-1} - \left(\gamma_5 - \frac{\gamma_6 (l_1 - 1) \varrho}{q + l_1 - 1} \right) |y|^{q+l_1-1}. \end{aligned} \quad (36)$$

Now we present some assumptions about the controller. The purpose of the Assumption 3.2 is to cope with the highly nonlinear nature of the underlying PSDE and estimate the upper bound of the sampling interval. Assumption 3.2 acts as a bridge between the infinitesimal operators of Lyapunov functions without proportional delays and sampling delays, and those with proportional delays and sampling delays.

Assumption 3.2 Under Hypothesis 1 and Hypothesis 2, for any controller $u : \mathbb{R}^n \times S \times \mathbb{R}_+ \rightarrow \mathbb{R}^n$ satisfying (5), there exist constants $\rho_j > 0$ ($1 \leq j \leq 8$) with $\rho_4 > \rho_5$ as well as $\rho_6 \in (0, 1)$, and a function $\Psi \in C(\mathbb{R}^n; \mathbb{R}_+)$ so that for $\forall (y, \hat{y}, i, t) \in \mathbb{R}^n \times \mathbb{R}^n \times S \times \mathbb{R}_+$

$$\begin{aligned} & \mathcal{L}U(y, \hat{y}, i, t) + \rho_1(2\vartheta_i|y| + (l_1 + 1)\bar{\vartheta}_i|y|^{l_1})^2 + \rho_2|f(y, \hat{y}, i, t)|^2 + \rho_3|g(y, \hat{y}, i, t)|^2 \\ & \leq -\rho_4|y|^2 + \rho_5\varrho|\hat{y}|^2 - \Psi(y) + \rho_6\varrho\Psi(\hat{y}), \end{aligned} \quad (37)$$

and

$$\rho_7|y|^{q+l_1-1} \leq \Psi(y) \leq \rho_8(1 + |y|^{q+l_1-1}). \quad (38)$$

This lemma can be proved in the same way as [4, 6] so the proof is omitted.

Remark 2 According to the result in [4], we can know that the control set that satisfies the Lipschitz condition (5), Assumption 3.1 and Assumption 3.2 is not an empty set.

We will present some theorems pertaining to the stabilization by the sampled-data control.

Theorem 3.4 Let the Lipschitz condition (5), Hypothesis 1 and Hypothesis 2 along with Assumption 3.1 and Assumption 3.2 hold. Furthermore if τ satisfies

$$\tau < \frac{\sqrt{(\rho_4 - \rho_5)\rho_1}}{2l^2} \wedge \frac{\sqrt{\rho_1\rho_2}}{\sqrt{2}l} \wedge \frac{\rho_1\rho_3}{l^2} \wedge \frac{1}{4\sqrt{2}l}, \quad (39)$$

then the solution of the PSDE given by (4) and initialized with (10) meets

$$\limsup_{t \rightarrow \infty} \frac{\log \mathbb{E}|y(t)|^2}{\log(1+t)} < 0. \quad (40)$$

Proof The proof is divided into three steps.

Step 1. In this proof the Lyapunov functional we use is

$$V(y(t), \sigma(t), t) = U(y(t), \sigma(t)) + I(t) \quad (41)$$

for $t \geq t_0$, where

$$I(t) = \frac{t^2}{\rho_1} \int_{-\tau}^0 \int_{t+s}^t \left[\tau |f_\lambda + u_\lambda|^2 + |g_\lambda|^2 \right] d\lambda ds. \quad (42)$$

To simplify the notation, we let $f_v = f(y(v), y(\varrho v), r(v), v)$, $u_v = u(y(v - \pi(v)), r(v), v)$, $g_v = g(y(v), y(\varrho v), r(v), v)$ ($v \geq t_0$). Let ε be a sufficiently small positive number which will be determined later. Apply generalised Itô formula [8] to $V(y(t \wedge \xi_k), \sigma(t), t)$, where ξ_k is given by (16). Since the exponential of each term $|y(t \wedge \xi_k)|$ is less than r by Hypothesis 2 and

Assumption 3.1, they are uniformly integrable. Then we get that

$$(1+t)^\varepsilon \mathbb{E}V(y(t), \sigma(t), t) \leq (1+t_0)^\varepsilon \mathbb{E}V(y(t_0), \sigma(t_0), t_0) + \int_{t_0}^t \mathbb{E}[\varepsilon(1+s)^{\varepsilon-1} V(y(s), r(s), s) + (1+s)^\varepsilon \mathbb{L}V(y(s), r(s), s)] ds, \quad (43)$$

by letting $k \rightarrow \infty$, where

$$\begin{aligned} \mathbb{L}V(y(t), \sigma(t), t) &= \mathcal{L}U_1(y(t), y(\varrho t), \sigma(t), t) + \frac{\iota^2}{\rho_1} \tau [\tau |f_t + u_t|^2 + |g_t|^2] \\ &\quad - \frac{\iota^2}{\rho_1} \int_{t-\tau}^t [\tau |f_\lambda + u_\lambda|^2 + |g_\lambda|^2] d\lambda, \end{aligned} \quad (44)$$

and

$$\begin{aligned} \mathcal{L}U_1(y(t), y(\varrho t), i, t) &= 2\vartheta_i \left[y^T [f(y(t), y(\varrho t), i, t) + u(y(t - \pi t), i, t)] + \frac{1}{2} |g(y(t), y(\varrho t), i, t)|^2 \right] \\ &\quad + (l_1 + 1) \bar{\vartheta}_i \left[|y(t)|^{l_1-1} y(t)^T [f(y(t), y(\varrho t), i, t) + u(y(t - \pi t), i, t)] \right. \\ &\quad \left. + \frac{1}{2} |y(t)|^{l_1-1} |g(y(t), y(\varrho t), i, t)|^2 + \frac{l_1-1}{2} |y(t)|^{l_1-3} |y(t)^T g(y(t), y(\varrho t), i, t)|^2 \right] \\ &\quad + \sum_{j=1}^N \gamma_{i,j} (\vartheta_j |y(t)|^2 + \bar{\vartheta}_j |y(t)|^{l_1+1}), \end{aligned} \quad (45)$$

By comparing (45) with (33), we can obtain

$$\begin{aligned} \mathbb{L}V(y(t), \sigma(t), t) &= \mathcal{L}U(y(t), y(\varrho t), \sigma(t), t) \\ &\quad + \frac{\iota^2}{\rho_1} \tau [\tau |f_t + u_t|^2 + |g_t|^2] - \frac{\iota^2}{\rho_1} \int_{t-\tau}^t [\tau |f_\lambda + u_\lambda|^2 + |g_\lambda|^2] d\lambda \\ &\quad + [2\vartheta_{\sigma(t)} + (l_1 + 1) \bar{\vartheta}_{\sigma(t)} |y(t)|^{l_1-1}] y^T(t) [u(y(t - \pi(t)), \sigma(t), t) - u(y(t), \sigma(t), t)], \end{aligned} \quad (46)$$

where $\mathcal{L}U$ is defined by (33). On the other hand,

$$\begin{aligned} &\mathbb{E} \mathbb{L}V(y(t), \sigma(t), t) \\ &\leq 2\vartheta_i \mathbb{E} \left[y^T(t) [f_t + u_t] + \frac{1}{2} |g_t|^2 \right] + \sum_{j=1}^N \gamma_{\sigma(t)j} (\vartheta_j \mathbb{E} |y(t)|^2 + \bar{\vartheta}_j \mathbb{E} |y(t)|^{l_1+1}) \\ &\quad + (l_1 + 1) \bar{\vartheta}_i \mathbb{E} \left[|y(t)|^{l_1-1} (y^T(t) [f_t + u_t] + \frac{l_1}{2} |g_t|^2) \right] \\ &\quad + \frac{\iota^2}{\rho_1} \tau [\tau \mathbb{E} |f_t + u_t|^2 + \mathbb{E} |g_t|^2] - \frac{\iota^2}{\rho_1} \mathbb{E} \int_{t-\tau}^t [\tau |f_\lambda + u_\lambda|^2 + |g_\lambda|^2] d\lambda \\ &\quad + \mathbb{E} \left([2\vartheta_{\sigma(t)} |y(t)| + (l_1 + 1) \bar{\vartheta}_{\sigma(t)} |y(t)|^{l_1}] |y(t - \pi(t)) - y(t)| \right). \end{aligned}$$

By (5) and Hypotheses 1 as well as Lemma 2.1 and Assumption 3.1, we can get

$$\mathbb{E} \mathbb{L}V(y(t), \sigma(t), t) < \infty, \quad \forall t \geq t_0. \quad (47)$$

Step 2. Let us estimate $\mathbb{L}V(y(t), \sigma(t), t)$. Firstly, Let

$$a_1 = \min_{i \in S} \vartheta_i, \quad a_2 = \max_{i \in S} \vartheta_i, \quad a_3 = \max_{i \in S} \bar{\vartheta}_i,$$

we get

$$\begin{aligned} a_1(1+t)^\varepsilon \mathbb{E}|y(t)|^2 &\leq (1+t_0)^\varepsilon \mathbb{E}V(y(t_0), \sigma(t_0), t_0) + \frac{\varepsilon \iota^2}{\rho_1} J_1(t) \\ &+ \int_{t_0}^t (1+s)^\varepsilon [\varepsilon a_2 \mathbb{E}|y(s)|^2 + \varepsilon a_3 \mathbb{E}|y(s)|^{l_1+1} + \mathbb{E}\mathbb{L}V(y(s), r(s), s)] ds, \end{aligned} \quad (48)$$

where

$$J_1(t) = \mathbb{E} \int_{t_0}^t (1+s)^{\varepsilon-1} \left(\int_{-\tau}^0 \int_{s+u}^s [\tau |f_\lambda + u_\lambda|^2 + |g_\lambda|^2] d\lambda du \right) ds.$$

It follows from (5) that

$$\begin{aligned} &[2\vartheta_{\sigma(t)} + (l_1 + 1)\bar{\vartheta}_{\sigma(t)} |y(t)|^{l_1-1}] y^T(t) [u(y(t - \pi(t)), \sigma(t), t) - u(y(t), \sigma(t), t)] \\ &\leq \rho_1 [2\vartheta_{\sigma(t)} |y(t)| + (l_1 + 1)\bar{\vartheta}_{\sigma(t)} |y(t)|^{l_1}]^2 + \frac{\iota^2}{4\rho_1} |y(t) - y(t - \pi(t))|^2. \end{aligned} \quad (49)$$

Next we observe from (39) that

$$\frac{2\iota^2 \tau^2}{\rho_1} \leq \rho_2, \quad \frac{\iota^2}{\tau} \leq \rho_3. \quad (50)$$

It then follows from (46) along with (5) and Assumption 3.2 that

$$\begin{aligned} \mathbb{L}V(y(t), \sigma(t), t) &\leq -\rho_4 |y(t)|^2 + \rho_5 \varrho |y(\varrho t)|^2 - \Psi(y(t)) + \rho_6 \varrho \Psi(y(\varrho t)) \\ &+ \frac{2\iota^4 \tau^2}{\rho_1} |y(t - \pi(t))|^2 + \frac{\iota^2}{4\rho_1} |y(t) - y(t - \pi(t))|^2 \\ &- \frac{\iota^2}{\rho_1} \int_{t-\tau}^t [\tau |f_\lambda + u_\lambda|^2 + |g_\lambda|^2] d\lambda. \end{aligned} \quad (51)$$

Noting that $\iota\tau \leq 1/4\sqrt{2}$, we have

$$\frac{2\iota^4 \tau^2}{\rho_1} |y(t - \pi(t))|^2 \leq \frac{4\iota^4 \tau^2}{\rho_1} |y(t)|^2 + \frac{\iota^2}{8\rho_1} |y(t) - y(t - \pi(t))|^2. \quad (52)$$

Finally, taking the expectation on both sides of (51), and then combining with (52), we get

$$\begin{aligned} \mathbb{E}\mathbb{L}V(y(t), \sigma(t), t) &\leq -\left(\rho_4 - \frac{4\iota^4 \tau^2}{\rho_1}\right) \mathbb{E}|y(t)|^2 + \rho_5 \varrho \mathbb{E}|y(\varrho t)|^2 - \mathbb{E}\Phi(y(t)) \\ &+ \rho_6 \varrho \mathbb{E}\Phi(y(\varrho t)) + \frac{3\iota^2}{8\rho_1} \mathbb{E}|y(t) - y(t - \pi(t))|^2 \\ &- \frac{\iota^2}{\rho_1} \mathbb{E} \int_{t-\tau}^t [\tau |f_\lambda + u_\lambda|^2 + |g_\lambda|^2] d\lambda. \end{aligned} \quad (53)$$

On the other hand, it follows from the PSDE (4) that

$$\begin{aligned}
\mathbb{E}|y(t) - y(t - \pi(t))|^2 &= \mathbb{E} \left| \int_{t-\pi(t)}^t [f_\lambda + u_\lambda] d\lambda + \int_{t-\pi(t)}^t g_\lambda dB(\lambda) \right|^2 \\
&\leq 2\mathbb{E} \int_{t-\pi(t)}^t (\tau |f_\lambda + u_\lambda|^2 + |g_\lambda|^2) d\lambda \\
&\leq 2\mathbb{E} \int_{t-\tau}^t (\tau |f_\lambda + u_\lambda|^2 + |g_\lambda|^2) d\lambda.
\end{aligned} \tag{54}$$

Substituting (54) into (53), then

$$\begin{aligned}
\mathbb{E}LV(y(t), \sigma(t), t) &\leq -\left(\rho_4 - \frac{4\iota^4\tau^2}{\rho_1}\right) \mathbb{E}|y(t)|^2 + \rho_5 \varrho \mathbb{E}|y(\varrho t)|^2 - \mathbb{E}\Psi(y(t)) \\
&\quad + \rho_6 \varrho \mathbb{E}\Psi(y(\varrho t)) - \frac{\iota^2}{4\rho_1} \mathbb{E} \int_{t-\tau}^t [\tau |f_\lambda + u_\lambda|^2 + |g_\lambda|^2] d\lambda.
\end{aligned} \tag{55}$$

Step 3. Furthermore, it is evident that

$$\mathbb{E}|y(t)|^{\iota_1+1} \leq \mathbb{E}|y(t)|^2 + \frac{1}{\rho_7} \mathbb{E}\Psi(y(t)). \tag{56}$$

By substituting (55) and (56) into (48), we have

$$\begin{aligned}
&a_1(1+t)^\varepsilon \mathbb{E}|y(t)|^2 \\
&\leq (1+t_0)^\varepsilon \mathbb{E}V(y(t_0), \sigma(t_0), t_0) + \frac{\varepsilon \iota^2}{\rho_1} J_1(t) - \frac{\iota^2}{4\rho_1} J_2(t) \\
&\quad - \left(\rho_4 - \frac{4\iota^4\tau^2}{\rho_1} - \varepsilon(a_2 + a_3)\right) \int_{t_0}^t (1+s)^\varepsilon \mathbb{E}|y(s)|^2 ds + \rho_5 \varrho \int_{t_0}^t (1+s)^\varepsilon \mathbb{E}|y(\varrho s)|^2 ds \\
&\quad - \left(1 - \frac{\varepsilon a_3}{\rho_7}\right) \int_{t_0}^t (1+s)^\varepsilon \mathbb{E}\Psi(y(s)) ds + \rho_6 \varrho \int_{t_0}^t (1+s)^\varepsilon \mathbb{E}\Psi(y(\varrho s)) ds,
\end{aligned} \tag{57}$$

where

$$J_2(t) = \mathbb{E} \int_{t_0}^t (1+s)^\varepsilon \left(\int_{s-\tau}^s [\tau |f_\lambda + u_\lambda|^2 + |g_\lambda|^2] d\lambda \right) ds. \tag{58}$$

Meanwhile, we can get

$$J_1(t) \leq \tau J_2(t), \tag{59}$$

$$\begin{aligned}
&\rho_5 \varrho \int_{t_0}^t (1+s)^\varepsilon \mathbb{E}|y(\varrho s)|^2 ds \\
&\leq \varrho^{-\varepsilon} \rho_5 \left(\int_{\varrho t_0}^{t_0} (1+s)^\varepsilon \mathbb{E}|y(s)|^2 ds + \int_{t_0}^t (1+s)^\varepsilon \mathbb{E}|y(s)|^2 ds \right)
\end{aligned} \tag{60}$$

and

$$\begin{aligned}
&\rho_6 \varrho \int_{t_0}^t (1+s)^\varepsilon \mathbb{E}\Psi(y(\varrho s)) ds \\
&\leq \varrho^{-\varepsilon} \rho_6 \left(\int_{\varrho t_0}^{t_0} (1+s)^\varepsilon \mathbb{E}\Psi(y(s)) ds + \int_{t_0}^t (1+s)^\varepsilon \mathbb{E}\Psi(y(s)) ds \right).
\end{aligned} \tag{61}$$

Now, we will select a sufficient small ε so that

$$\varepsilon\tau \leq \frac{1}{4}, \quad \varepsilon(a_2 + a_3) + \rho_5\varrho^{-\varepsilon} \leq \rho_4 - \frac{4\ell^4\tau^2}{\rho_1}, \quad \rho_6\varrho^{-\varepsilon} + \frac{\varepsilon a_3}{\rho_7} \leq 1.$$

Substituting (59)-(61) into (57), then

$$\mathbb{E}|y(t)|^2 \leq C_2(1+t)^{-\varepsilon}, \quad (62)$$

where

$$\begin{aligned} C_2 = & a_1^{-1}[(1+t_0)^\varepsilon \mathbb{E}V(y(t_0), \sigma(t_0), t_0) + \varrho^{-\varepsilon} \rho_5 \int_{\varrho t_0}^{t_0} (1+s)^\varepsilon \mathbb{E}|y(s)|^2 ds \\ & + \varrho^{-\varepsilon} \rho_6 \int_{\varrho t_0}^{t_0} (1+s)^\varepsilon \mathbb{E}\Psi(y(s)) ds]. \end{aligned}$$

Remark 3 Theorem 3.4 not only shows that it is possible to design a control function for the controlled system (4) to become polynomial stabilization in L^2 , but also provides an estimate of the upper bound of the sampling interval τ . It should be pointed out that we can make τ more suitable by adjusting the parameters ρ_j ($1 \leq j \leq 8$) in Assumption 3.2.

In the following, we show that it is also possible to make the controlled system polynomially stable in $L^{\bar{r}}$ for some $\bar{r} > 2$.

Theorem 3.5 *We replace Hypothesis 2 with Hypothesis 3 in Theorem 3.4. then the solution of the PSDE given by (4) and initialized with (10) meets*

$$\limsup_{t \rightarrow \infty} \frac{\log \mathbb{E}|y(t)|^{\bar{r}}}{\log(1+t)} < 0, \quad \forall \bar{r} \in [2, r). \quad (63)$$

Proof Fix any $\bar{r} \in (2, r)$. For a constant $v = (r - \bar{r})/(r - 2) \in (0, 1)$, the Hölder inequality shows

$$\mathbb{E}|y(t)|^{\bar{r}} = \mathbb{E}(|y(t)|^{2v} |y(t)|^{\bar{r}-2v}) \leq (\mathbb{E}|y(t)|^2)^v (\mathbb{E}|y(t)|^{(\bar{r}-2v)/(1-v)})^{1-v}.$$

it is easy to show that

$$\begin{aligned} \mathbb{E}|y(t)|^{\bar{r}} & \leq (\mathbb{E}|y(t)|^2)^{(r-\bar{r})/(r-2)} (\mathbb{E}|y(t)|^r)^{(\bar{r}-2)/(r-2)} \\ & \leq C_3^{(\bar{r}-2)/(r-2)} (\mathbb{E}|y(t)|^2)^{(r-\bar{r})/(r-2)}. \end{aligned} \quad (64)$$

where $C_3 =: \sup_{t_0 \leq t < \infty} \mathbb{E}|y(t)|^r$. Combining (62) and (64), then

$$\mathbb{E}|y(t)|^{\bar{r}} \leq C_4(1+t)^{-\varepsilon v} \quad (65)$$

for all $t \geq t_0$, where $C_4 = C_3^{(\bar{r}-2)/(r-2)} C_2^{(r-\bar{r})/(r-2)}$. According to (65), we can deduce the assertion (63).

In general, it is not possible to imply the almost surely polynomial stabilization from the $\bar{r}$ th moment polynomial stabilization. However, in our situation, this is possible as described in the following theorem.

Theorem 3.6 *If all the requirements of Theorem 3.5 can be satisfied, then the solution of the PSDE given by (4) and initialized with (10) meets*

$$\limsup_{t \rightarrow \infty} \frac{\log |y(t)|}{\log(1+t)} < 0 \quad a.s. \quad (66)$$

Proof Let $t_k = t_0 + k\tau$ for $k = 0, 1, 2, \dots$. By the Hölder inequality and the Doob martingale inequality^[8], we can show that

$$\begin{aligned} \mathbb{E}\left(\sup_{t_k \leq t \leq t_{k+1}} |y(t)|^2\right) &\leq 3\tau \mathbb{E} \int_{t_k}^{t_{k+1}} (|f(y(t), y(\varrho t), \sigma(t), t) + u(y(t - \pi(t)), \sigma(t), t)|^2) \\ &\quad + 3\mathbb{E}|y(t_k)|^2 + 12\mathbb{E} \int_{t_k}^{t_{k+1}} |g(y(t), y(\varrho t), \sigma(t), t)|^2 dt. \end{aligned} \quad (67)$$

Let $\bar{r} = 2(l_1 \vee l_2 \vee l_3 \vee l_4)$. Recalling (7), (8) and $l_1 > 1$, we see $\bar{r} \in [2, r)$. By Hypothesis 1, it is almost straightforward to show from (67) that

$$\begin{aligned} \mathbb{E}\left(\sup_{t_k \leq t \leq t_{k+1}} |y(t)|^2\right) &\leq 3\mathbb{E}|y(t_k)|^2 \\ &\quad + C_5 \int_{t_k}^{t_{k+1}} \mathbb{E}(|y(t)|^2 + |y(\varrho t)|^2 + |y(t - \pi(t))|^2 + |y(t)|^{\bar{r}} + |y(\varrho t)|^{\bar{r}}) dt. \end{aligned}$$

where C_5 and the following C_6 are all positive constants independent of k . Using (62) and (65), we hence have

$$\mathbb{E}\left(\sup_{t_k \leq t \leq t_{k+1}} |y(t)|^2\right) \leq C_6(1+t_k)^{-\varepsilon^*},$$

where $\varepsilon^* = \frac{\varepsilon(r-\bar{r})}{r-2}$. Consequently

$$\sum_{k=0}^{\infty} P\left(\sup_{t_k \leq t \leq t_{k+1}} |y(t)| > (1+t_k)^{-0.25\varepsilon^*}\right) \leq \sum_{k=0}^{\infty} C_6(1+t_k)^{-0.5\varepsilon^*} < \infty.$$

According to the Borel-Cantelli lemma, it shows that for almost all $\omega \in \Omega$, there exists a positive integer $k_0 = k_0(\omega)$ such that

$$\sup_{t_k \leq t \leq t_{k+1}} |y(t)| \leq (1+t_k)^{-0.25\varepsilon^*}, \quad k \geq k_0.$$

So we have

$$\frac{\log(|y(t)|)}{\log(1+t)} \leq \frac{-0.25\varepsilon^* \log(1+t_k)}{\log(1+t)}, \quad t \in [t_k, t_{k+1}], \quad k \geq k_0,$$

which implies

$$\limsup_{t \rightarrow \infty} \frac{\log |y(t)|}{\log(1+t)} < 0 \quad a.s.$$

This concludes the proof.

4 Numerical example

Example. To facilitate the computations, we focus on a scalar highly nonlinear PSDE with 2-state Markovian switching described by the equation

$$dy(t) = f(y(t), y(0.9t), \sigma(t), t)dt + g(y(t), y(0.9t), \sigma(t), t)dB(t) \quad (68)$$

initialized with the provided data $\zeta(t) = y_0(0.9 \leq t \leq 1)$ and $r(1) = 1$. In this case, the definitions of f and g are as follows

$$\begin{aligned} f(y, \hat{y}, 1, t) &= y(1 - 5y^2 + 0.5\hat{y}^2), & g(y, \hat{y}, 1, t) &= |y|^{3/2} + 0.5\hat{y}, \\ f(y, \hat{y}, 2, t) &= y(1 - 4y^2 - 0.5\hat{y}^2), & g(y, \hat{y}, 2, t) &= 0.5|y|^{3/2} - 0.5\hat{y}, \end{aligned}$$

for all $y, \hat{y} \in \mathbb{R}$, $\Gamma = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$ is the generator of $\sigma(t)$. We can confirm that Hypotheses 1 and 2 are satisfied when $q = 4$, $l_1 = l_2 = 3$, $l_3 = 1.5$, $l_4 = 1$, $\alpha_1 = [1 + 0.25(r - 1)^2] \vee (r - 1)$, $\alpha_2 = 0.28(r - 1)$, $\alpha_3 = 3.5$, $\alpha_4 = 0.28$ and for any $r > 6$. We can observe that the hybrid PSDE (68) exhibits instability. This finding can be corroborated by referring to the Figure 1 and the Figure 2.

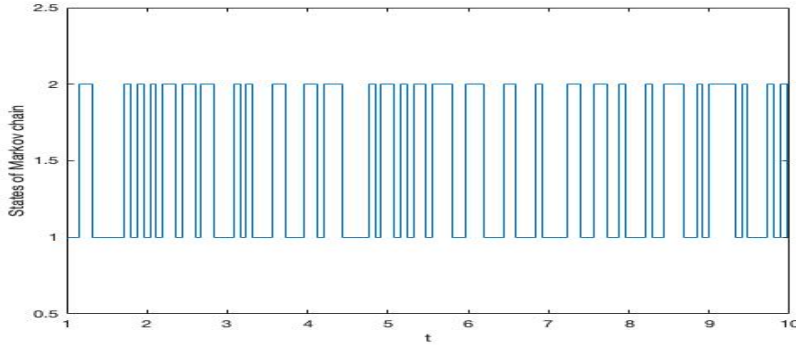


Figure 1: Markov chain.

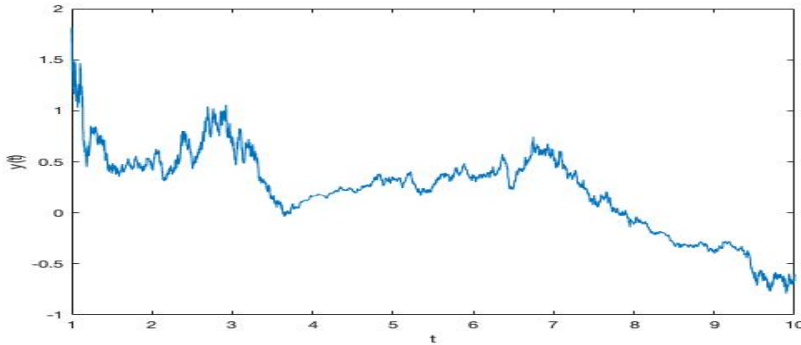


Figure 2: The state trajectory of the PSDE (68)

Figure 1 shows the state transition of two state Markov chains, and Figure 2 shows the solution of the PSDE (68) using the Euler–Maruyama technique (The step size is 10^{-3}).

To stabilize the PSDE (68), we employ a sampled-data controller to construct the system with controller

$$\begin{aligned} dy(t) = & [f(y(t), y(0.9t), \sigma(t), t) + u(y(\delta_t), \sigma(t), t)]dt \\ & + g(y(t), y(0.9t), \sigma(t), t)dB(t) \end{aligned} \quad (69)$$

where

$$u(y, 1, t) = -5y; \quad u(y, 2, t) = -4y. \quad (70)$$

We can verify that (5) holds when $\iota = 5$. By Lemma 2.1, the solution $y(t)$ of PSDE (69) satisfies (11). Then, we will check Assumption 3.1. We can find that

$$\begin{aligned} & y^T [f(y, \hat{y}, i, t) + u(y, i, t)] + \frac{1}{2} |g(y, \hat{y}, i, t)|^2 \\ & \leq \begin{cases} -3.5y^2 + 0.278 \times 0.9\hat{y}^2 - 4.25y^4 + 0.278 \times 0.9\hat{y}^4, & i = 1, \\ -2.875y^2 + 0.278 \times 0.9\hat{y}^2 - 4.125y^4 + 0.278 \times 0.9\hat{y}^4, & i = 2. \end{cases} \end{aligned}$$

$$\begin{aligned} & y^T [f(y, \hat{y}, i, t) + u(y, i, t)] + \frac{l_1}{2} |g(y, \hat{y}, i, t)|^2 \\ & \leq \begin{cases} -2.5y^2 + 0.834 \times 0.9\hat{y}^2 - 3.25y^4 + 0.834 \times 0.9\hat{y}^4, & i = 1, \\ -2.625y^2 + 0.834 \times 0.9\hat{y}^2 - 3.375y^4 + 0.834 \times 0.9\hat{y}^4, & i = 2. \end{cases} \end{aligned}$$

So (27)-(28) hold with

$$\begin{aligned} \tilde{a}_1 &= -3.5, & \tilde{b}_1 &= 0.278, & \tilde{c}_1 &= 4.25, & \tilde{d}_1 &= 0.278, \\ \tilde{a}_2 &= -2.875, & \tilde{b}_2 &= 0.278, & \tilde{c}_2 &= 4.125, & \tilde{d}_2 &= 0.278, \\ \hat{a}_1 &= -2.5, & \hat{b}_1 &= 0.834, & \hat{c}_1 &= 3.25, & \hat{d}_1 &= 0.834, \\ \hat{a}_2 &= -2.625, & \hat{b}_2 &= 0.834, & \hat{c}_2 &= 3.375, & \hat{d}_2 &= 0.834, \end{aligned}$$

and

$$M_1 = \begin{pmatrix} 9 & -2 \\ -2 & 7.75 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 12 & -2 \\ -2 & 12.5 \end{pmatrix},$$

which are both M-matrices. According to (31), we have

$$\vartheta_1 = 0.1483, \quad \vartheta_2 = 0.1673, \quad \bar{\vartheta}_1 = 0.0993, \quad \bar{\vartheta}_2 = 0.0959.$$

Accordingly,

$$\begin{aligned} \gamma_1 &= 0.0930, & \gamma_2 &= 1.2605, & \gamma_3 &= 0.0930, \\ \gamma_4 &= 0.3313, & \gamma_5 &= 1.2911, & \gamma_6 &= 0.3313, \end{aligned}$$

which meet (30). That is to say, the control function $u(y, i)$ adheres to Assumption 3.1. Moreover, it is evident that

$$U(y, i) = \begin{cases} 0.1483y^2 + 0.0993y^4, & i = 1, \\ 0.1673y^2 + 0.0959y^4, & i = 2. \end{cases}$$

According to (36), the following inequality holds,

$$\mathcal{L}U(y, \hat{y}, i, t) \leq -y^2 + 0.0837\hat{y}^2 - 2.1114y^4 + 0.2328\hat{y}^4 - 1.1917y^6 + 0.1918\hat{y}^6.$$

Simultaneously, we obtain

$$\begin{aligned} (2\vartheta_i|y| + (l_1 + 1)\bar{\vartheta}_i|y|^{l_1})^2 &\leq 0.1120y^2 + 0.2567y^4 + 0.1458y^6, \\ |f(y, \hat{y}, i, t)|^2 &\leq y^2 - 9y^4 + 0.5\hat{y}^4 + 25.0833y^6 + 1.6667\hat{y}^6, \\ |g(y, \hat{y}, i, t)|^2 &\leq 0.5y^2 + 0.5\hat{y}^2 + 2y^4. \end{aligned}$$

Choosing $\rho_1 = 0.6$, $\rho_2 = 0.01$ and $\rho_3 = 0.5$, we get

$$\begin{aligned} \mathcal{L}U(y, \hat{y}, i, t) + \rho_1(2\vartheta_i|y| + (l_1 + 1)\bar{\vartheta}_i|y|^{l_1})^2 + \rho_2|f(y, \hat{y}, i, t)|^2 + \rho_3|g(y, \hat{y}, i, t)|^2 \\ \leq -0.6728y^2 + 0.3708 \times 0.9000\hat{y}^2 - \Psi(y) + 0.2807 \times 0.9000\Psi(\hat{y}), \end{aligned}$$

where $\Psi(y) = 1.0474y^4 + 0.8534y^6$, $\rho_4 = 0.6728$, $\rho_5 = 0.3708$, $\rho_6 = 0.2807$, $\rho_7 = 0.8534$ and $\rho_8 = 1.9008$. By (39), we deduce that the system (69) can achieve polynomial stabilization in L^2 with $\bar{\tau} < 0.0085$.

The process of computer simulation is implemented through the Euler-Maruyama technique (The step size is 10^{-3}). Let $\tau = 0.008$, $r(0) = 1$ and $y(t_0) = 1 + \sin(t_0)$, $t_0 \in [0.9, 1]$.

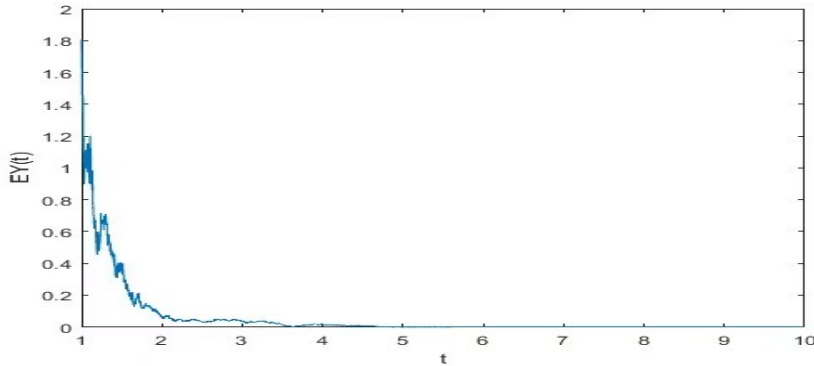


Figure 3: The state trajectory of the solution.

Figure 3 shows the time evolution of the solution of the PSDE (69).

5 Conclusions

In this paper, we have embarked on an in-depth exploration of the stabilization for a specific class of hybrid PSDEs. These PSDEs, while inherently unstable, exhibit finite or bounded solutions. This unique characteristic has been the focal point of our study.

Our analysis extended to the r th moment polynomial stability and almost surely polynomial stability, both of which were examined through the lens of a sampled-data control mechanism. This approach allowed us to delve into the intricate dynamics of these systems and understand their behavior under different conditions. To substantiate our theoretical findings, we provided an illustrative example accompanied by computer simulations. These practical applications serve to bridge the gap between theory and practice, demonstrating the real-world implications of our research.

Looking forward, we believe that our work lays a solid foundation for future research in this area. The methods and insights presented in this paper could be instrumental in furthering our understanding of PSDEs and their maneuvers. Moreover, the novel criterion we established for the boundedness of the state of PSDEs opens up new avenues for effective investigation.

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